Balla Veekshita Naidu

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Experience

Senior Software Engineer May 2025 – Present
Software Engineer May 2024 – May 2025
Dvara E-Registry — Python, Django, PostgreSQL, AWS *Hyderabad*

- **Owned production backend services** for a crop-insurance platform, covering REST API design, PostgreSQL data modelling, asynchronous processing, and AWS deployments.
- Built a **multi-stage transaction reconciliation and approval workflow** in Django, processing **∼10K transactions and repayments per month** and reducing reconciliation from hours of manual checks to minutes. Implemented **stage-specific validation, API orchestration, and a manual-review queue** for mismatched records, routing exceptions for human review.
- Designed **Celery-based task orchestration** with each reconciliation stage executed as a separate task, configurable retries for recoverable failures, and failure isolation to prevent bad records from blocking batch processing.
- Implemented **shared API quota enforcement** for services used by multiple internal teams, combining cached usage limits with counters distributed across **10 database rows** to reduce contention on a single hot row.
- Built **parallel geospatial processing pipelines** that partition thousands of grid cells into non-overlapping chunks across multiple workers, preventing duplicate processing during large backfills.
- Developed data pipelines to synchronize **geospatial parameters from external APIs and Google datasets** across platform database tables.
- Led development of a **Django + React farm-digitization platform**, automating farm-boundary capture through GPS-guided, landmark-based workflows and geospatial validation.
- Built a **FastAPI + React image-annotation platform** supporting approximately **20 concurrent annotators**, storing images and labels in AWS S3 and a database to produce a ground-truth dataset of approximately **7,000 images**.

AI/ML Intern Aug 2023 – Jan 2024
Dvara Solutions — Python, LLMs, Vector Databases, PostgreSQL, ASR *Dharwad*

- Built a **real-time voice-enabled RAG application** for querying domain-specific financial documents, integrating **Whisper** for ASR and **Pinecone/FAISS** for semantic vector search.
- Designed a document ingestion and **embedding-management pipeline** using vector databases and PostgreSQL metadata storage, supporting incremental updates and retrieval workflows.
- Implemented **multilingual ASR pipelines** for regional Indian languages using transcription and translation models, enabling cross-language voice interactions.

Technical Skills

Languages: Python, C++, Java, JavaScript, C, SQL
Backend & APIs: Django, Django REST Framework, FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices
Databases & Systems: PostgreSQL, MySQL, MongoDB, Database Design, Indexing, Query Optimization, Caching, Concurrency
Cloud & DevOps: AWS, Git, Linux
AI/ML & GenAI: LangChain, LangGraph, RAG, Pandas, TensorFlow, Scikit-learn, Vector Databases- FAISS, Pinecone
Computer Vision: YOLOv5, Faster R-CNN, ResNet, EfficientNet, Image Classification, Object Detection.
Frontend: React, HTML, CSS

Achievements

- Solved **700+ algorithmic problems** across LeetCode and GeeksforGeeks.
- Received the **Trailblazer Award** for outstanding contributions and impact at Dvara E-Registry.
- Received the **Top Performer Award** for consistently delivering high-impact engineering outcomes.
- Secured **4th position** in the Inter IIT Case Study Competition.

Projects

MyCV – Resume Builder | *Node.js, Express, MongoDB, React* | GitHub

- Built and deployed a full-stack resume builder with a **REST API backend**, profile-driven template autofill, PDF export, and persistent user data using MongoDB.

Animal Image Classification | *TensorFlow, Keras, Scikit-learn, NumPy, Pandas*

- Trained and compared VGG-16 (0.93 acc.) and ResNet50 (0.92 acc.) CNN models on **26,000 images across 10 categories**; benchmarked against a gradient boosting baseline (0.85 acc.), demonstrating a 9%+ lift with deep learning.

Education

Indian Institute of Technology, Dharwad (IIT Dh) 2020 – 2024
Bachelor of Technology (B.Tech) in Mechanical Engineering — CGPA: 9.09 / 10